ASSESSMENT

Saturday, November 1, 2025 7:36 PM

Assessment

Assessment Name	Weight	Qual Mark	Due Date	
Coursework	50	30	Wed, 11th Jan 1pm	Research, implement, and evaluate an AI application.
In-Class Test (On Blackboard)	50	30	Week 12 In your seminar (must attend in person or seek MC)	Multiple choice and short answer questions on the blackboard. Covering all topics up to 11. Duration 90 minutes.

2 - Agents and Environments

Saturday, November 1, 2025 7:49 PM

Here's a concise and well-organized **set of notes** based on your uploaded lecture, **"Applied AI – Lecture 2: Agents and Environments"**:

Applied AI – Lecture 2: Agents and Environments

Dr. Artie Basukoski

1. Intelligent Agents

- **Agent** = *Architecture + Program*
- **Definition:** An agent perceives its environment through **sensors** and acts upon it through **actuators**.
- **Agent Function:** maps percept sequences to actions → $f : P^* \rightarrow A$
- **Goal:** Create rational agents — do the "right thing" to maximize performance.

Characteristics of Intelligent Agents

- **Autonomy:** Operates without direct human control.
- **Reactivity:** Responds to environmental changes.
- **Proactiveness:** Takes initiative to achieve goals.
- **Social Ability:** Interacts with other agents or humans.

Examples

- Robotics, Software Agents, Virtual Assistants.

2. Rationality

- **Rational Agent:** For every percept sequence, selects an action that maximizes expected performance.
- **Depends on:**
  1. Performance measure (criteria of success)
  2. Prior knowledge of environment
  3. Actions available
  4. Percept history

  Rational ≠ Omniscient ≠ Clairvoyant ≠ Always Successful
  Rational ⇒ Exploration, Learning, Autonomy

3. PEAS Framework (Task Environment)

Used to design rational agents.

PEAS = Performance measure, Environment, Actuators, Sensors

Examples:

- **Automated Taxi:**
  - P: Safety, legality, comfort, profit
  - E: Roads, traffic, weather
  - A: Steering, brakes, horn
  - S: Cameras, GPS, sensors
- **Internet Shopping Agent:**
  - P: Price, quality, efficiency
  - E: Web pages, vendors
  - A: Click links, submit forms
  - S: Web content
- **ChatGPT Assistant Agent:**
  - P: Accuracy, relevance, user satisfaction
  - E: Web and databases
  - A: Display text, use plugins
  - S: Text, images, voice

4. Environment Types

Category	Description	Example
Fully Observable	Agent can access full state	Chess
Partially Observable	Limited perception	Poker
Deterministic	Next state predictable	Chess
Stochastic	Randomness involved	Stock market
Episodic	Independent episodes	Quiz show
Sequential	Current decisions affect future	Soccer
Static	Does not change while deliberating	Crossword
Dynamic	Changes while acting	Self-driving car
Discrete	Finite states/actions	Tic-tac-toe
Continuous	Infinite states/actions	Car steering
Single-Agent	One agent	Sudoku
Multi-Agent	Multiple agents	Traffic system

Real-world = partially observable, stochastic, sequential, dynamic, continuous, multi-agent.

Dimension	Characteristic	Definition	Example
Observability	**Fully Observable**	Agent can see all relevant aspects at once.	Chess
	Partially Observable	Agent cannot see everything relevant.	Poker (cannot see other players' cards)
Predictability	**Deterministic**	Next state is completely determined by current state and action.	Chess
	Stochastic	Next state is not entirely predictable.	Stock market
Sequence	**Episodic**	Agent's experience is divided into independent episodes.	Quiz game
	Sequential	Decisions are interdependent; one action affects future actions.	Soccer
Change Rate	**Static**	Environment does not change while agent is thinking.	Chess
	Dynamic	Environment changes even if the agent does nothing.	Self-driving car scenario
Granularity	**Discrete**	Finite number of actions and states.	Tic-Tac-Toe
	Continuous	Infinite range of possible states and actions.	Steering a car
Actors	**Single-Agent**	Only one agent is acting.	Sudoku solver
	Multi-Agent	Multiple interacting agents.	Stock trading system

5. Types of Agents

1. **Simple Reflex Agents:**
   - Condition-action rules; respond directly to percepts.
   - Example: Vacuum agent – if dirty → suck, else move.
2. **Model-Based Reflex Agents:**
   - Maintain internal state to handle partial observability.
3. **Goal-Based Agents:**
   - Act to achieve a goal (e.g., travel from Arad → Bucharest).
4. **Utility-Based Agents:**
   - Use utility function to evaluate outcomes.
   - Examples:
     - Autonomous vehicles: balance safety, time, fuel.
     - Trading systems: expected return – risk.
     - Recommendation systems: predicted rating – popularity.
5. **Learning Agents:**
   - **Performance Element:** Takes actions.
   - **Learning Element:** Improves behavior.
   - **Critic:** Evaluates actions.
   - **Problem Generator:** Suggests exploration.

     Examples:
   - *Amazon recommender*, *AlphaGo* (reinforcement learning).

6. Problem Formulation

Defined by:

1. **Initial state**
2. **Successor function (S(x))** – actions + resulting states
3. **Goal test**
4. **Path cost** (sum of step costs)

A solution: Sequence of actions from initial → goal state.

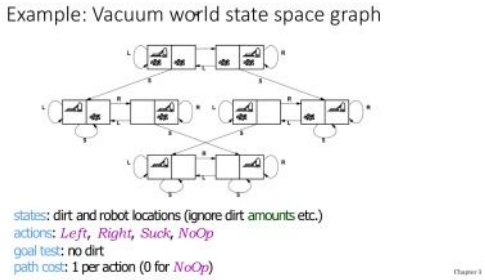
Examples:

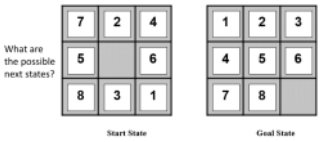
- **Romania Map Problem:** Path from Arad → Bucharest
- **Vacuum World:** Minimize dirt with minimal actions
- **8-Puzzle:** Arrange tiles to reach goal configuration
- **Robotic Assembly:** Continuous control for task completion

7. Problem Types

- **Single-state (deterministic, observable)**
  - Agent knows exactly which state it will be in; solution is a sequence
- **Sensorless (no observation)**
  - Agent may have no idea where it is; solution (if any) is a sequence
- **Contingency (partial observation)**
  - percepts provide new information about current state solution is a contingent plan or a policy often interleave search, execution
- **Exploration (unknown state space)**
  - exploration problem ("online")

8. Selecting a State Space

- 



states: integer locations of tiles (ignore intermediate positions)
actions: move blank left, right, up, down (ignore unjamming etc.)
goal test = goal state (given)
path cost: 1 per move

9. Current Research & Future Directions

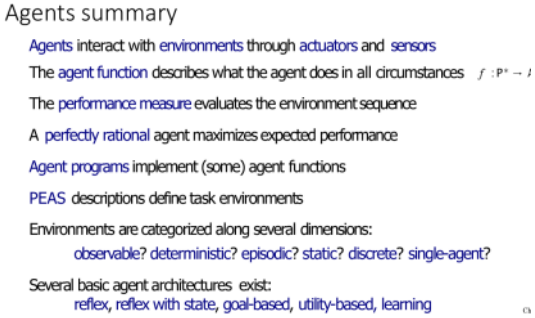
- **Focus Areas:** Multi-agent systems, human-agent interaction, learning.
- **Challenges:** Scalability, robustness, commonsense reasoning.
- **Applications:** Space exploration, nanotech, human augmentation.

Multi-Agent Systems

- **Applications:** Traffic control, smart grids, robotics, games.
- **Challenges:** Coordination, communication, learning in dynamic environments.

Ethical & Societal Implications

- Algorithmic bias and fairness
- Transparency and explainability
- Privacy & security
- Accountability and legal issues
- Workforce displacement



From <https://notebooklm.google.com/notebook/e2059a48-1d23-4d09-aa4a-4141d56b722b>

3 - Search strategies

Saturday, November 1, 2025 8:02 PM

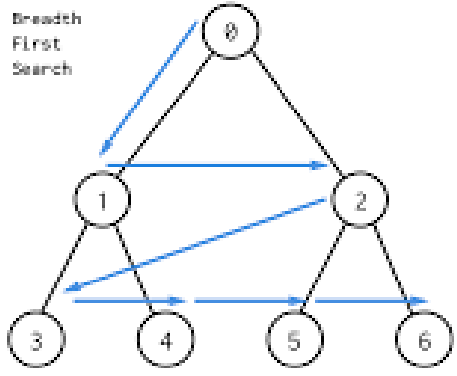
Covers the fundamental concepts of state-space search, starting with uninformed strategies that rely only on the problem definition, then moving to informed strategies that use heuristic knowledge to guide the search.

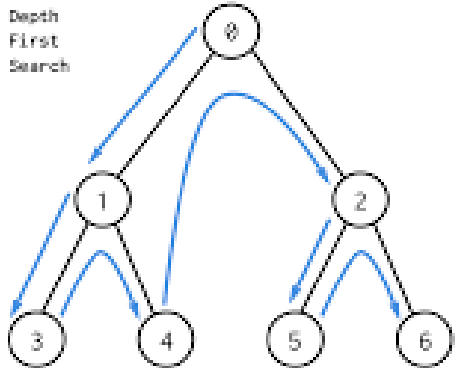
Agenda

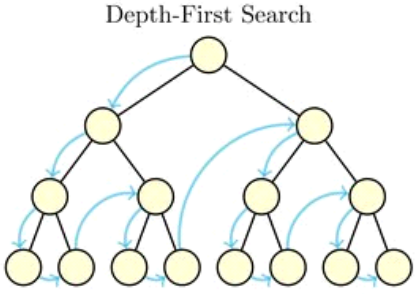
- Search
- Big Oh Notation
- Comparison of uninformed search algorithms
- Inform the search algorithms
- Comparison of informed search algorithms

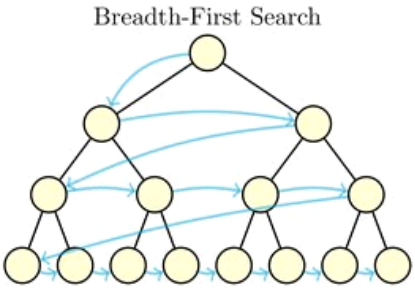
- **Search** is the process of exploring possible states to find a path from an **initial state** to a **goal state**.
- Each state represents a possible configuration.
- **Node:** data structure with:
  - State
  - Parent
  - Action
  - Path cost g(n)
  - Depth

Core idea:
Generate and expand nodes according to a strategy until a goal is found.





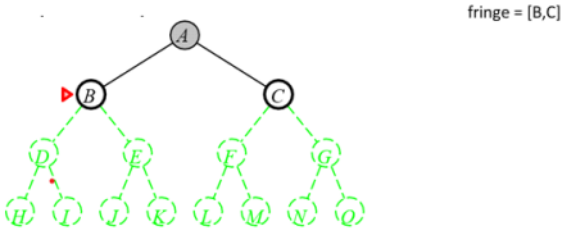




Depth First Search (STACK)

Expand deepest unexpanded node

Implementation:
fringe = LIFO queue, i.e., put successors at front

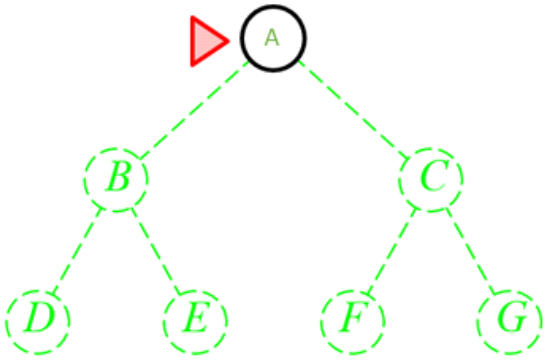


Order of the above tree search in DFS would be
A,B ,D ,H,I,E,J,K,C,F,L,M,G,N,O

Breadth First Search (QUEUE)

Expand shallowest unexpanded node

Implementation:
fringe is a FIFO queue, i.e., new successors go at end



Order of the above tree search in BFS would be
A,B,C,D,E,F,G

A strategy is defined by picking the order of node expansion

3. Key Evaluation Criteria for Search Strategies

Property	Description
Completeness	Guaranteed to find a solution if one exists
Time Complexity	Number of nodes generated/expanded
Space Complexity	Maximum nodes stored in memory
Optimality	Returns the least-cost solution

Uninformed search strategies

- Breadth-first search
- Uniform-cost search
- Depth-first search
- Depth-limited search
- Iterative deepening search

3. Depth-Limited Search (DLS)

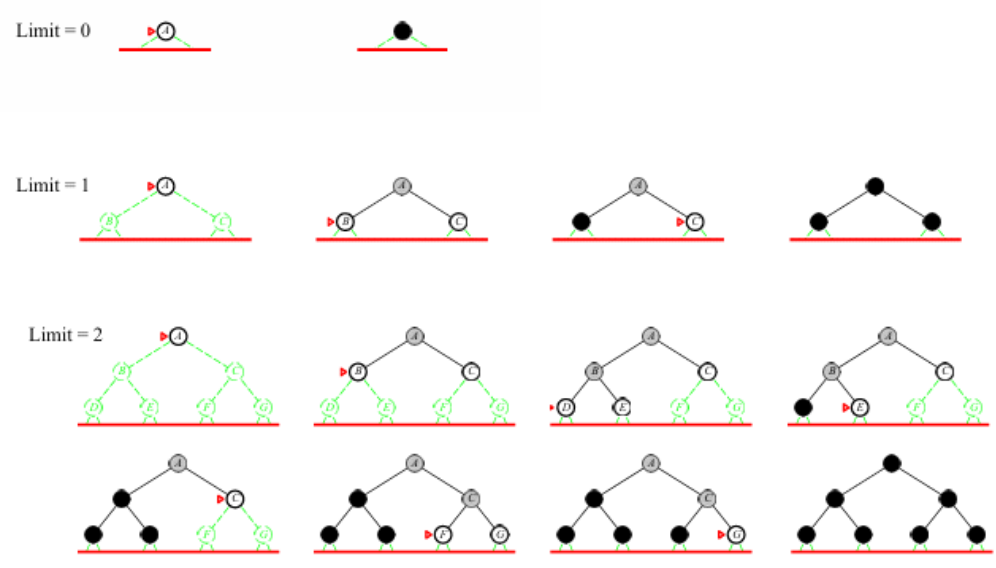
• **Concept:** A depth-first search variation that introduces a fixed **depth limit** *l*. Nodes at depth *l* are treated as having no successors.

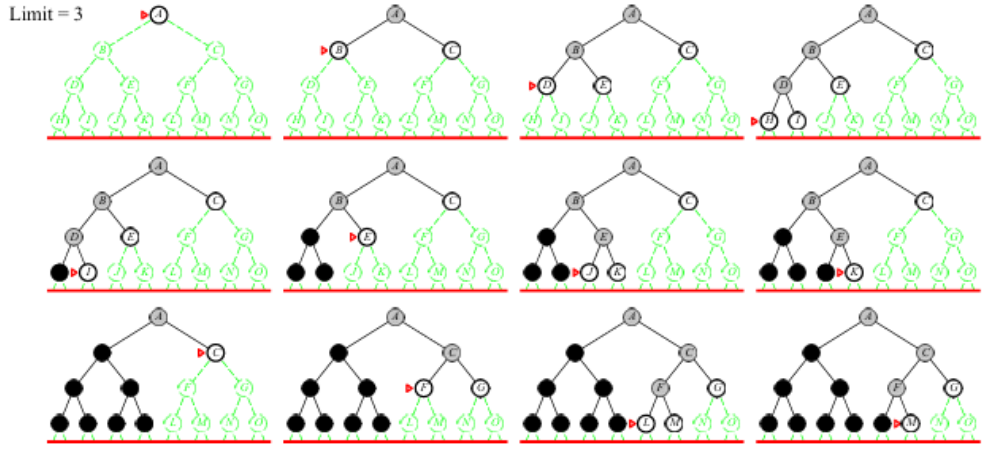
• **Implementation:** Can be implemented recursively

4. Iterative Deepening Search (IDS)

• **Concept:** Combines the benefits of BFS and DFS. It performs a sequence of depth-limited searches, where the depth limit increases from 0 up to ∞.

• **Process:** Starts with limit *l*=0, then *l*=1, *l*=2, and so on, until a solution is found





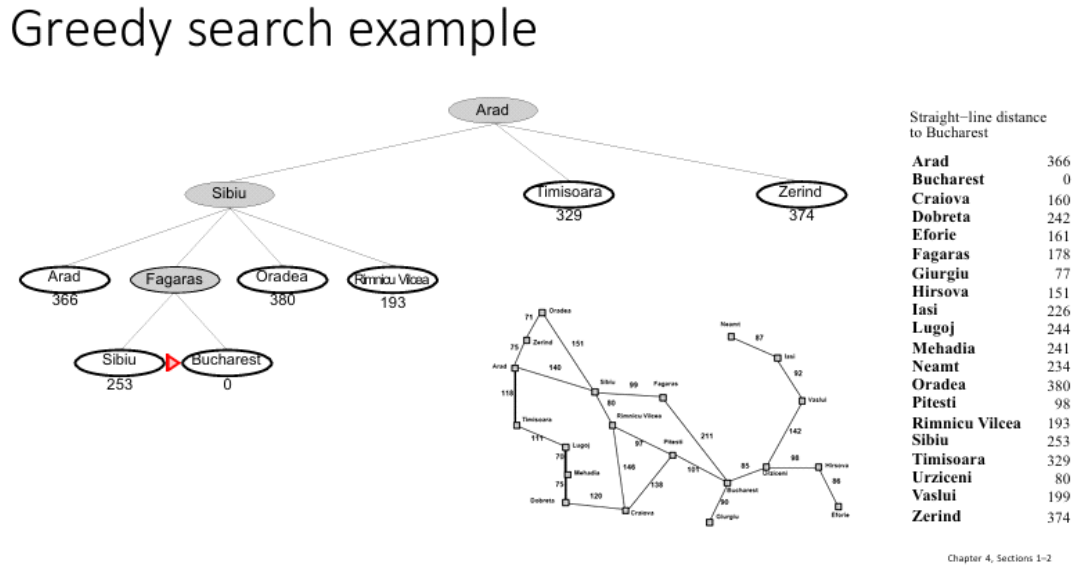
- **Depth-Limited Search (DLS)** may **fail** if the goal is **deeper than the chosen limit l**, because it stops searching beyond that depth — even if the goal exists there.
- **Iterative Deepening Search (IDS)** avoids this problem by **repeatedly running DLS with increasing limits** (l = 0, 1, 2, 3, …) until the **goal node is actually reached**.
  So in terms of finding the goal:
  IDS **guarantees that the goal node will be found**, no matter how deep it is (as long as it exists), while DLS might **miss the goal** if the limit is set too low.

Informed Search Algorithms

Use **domain knowledge** to estimate how close a node is to a goal.

1-Greedy Search

Greedy search expands the node that appears to be closest to goal



If node B is the goal node from node A.
Expand the available nodes to choose from and choose the lowest value.

In terms:
From A, the nodes with distances are as below
S 253
T 329
Z 374 , since S is the smallest distance we move to Node S from A

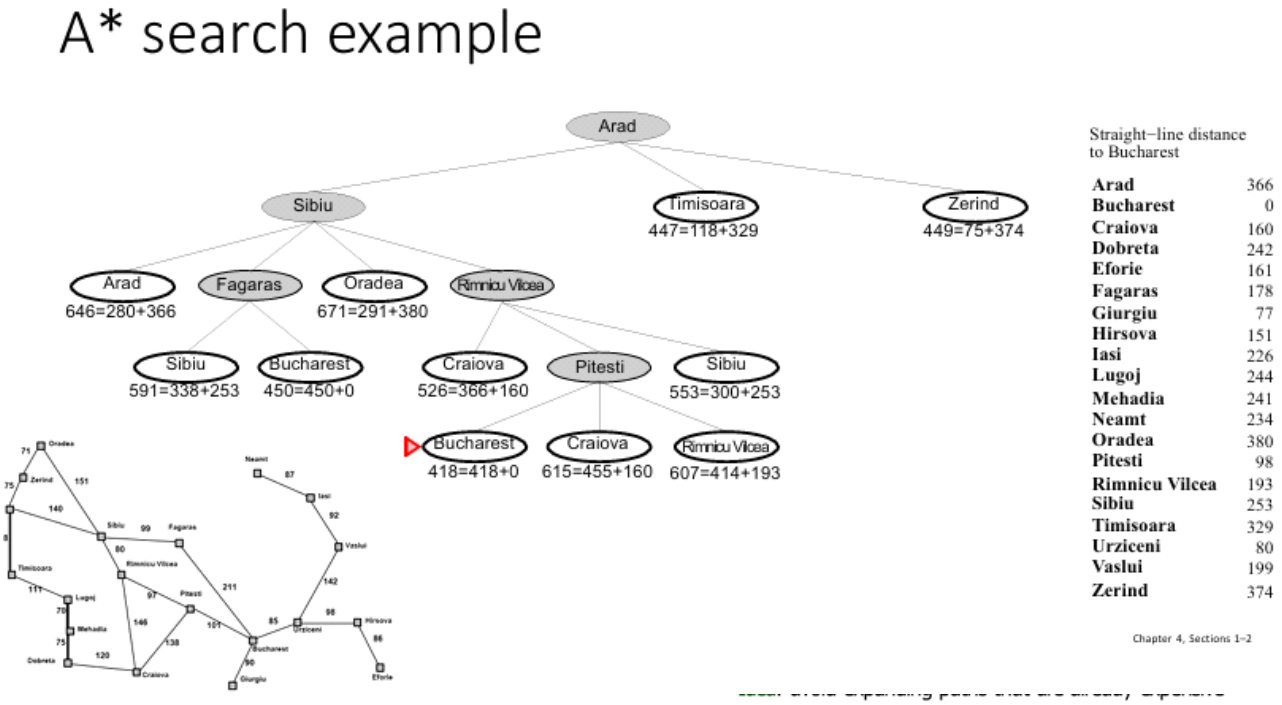
From S, the nodes with distances are as below

A 366
F 178
O 380
R 193 , since F is the shortest distance we move to Node F from S

Like so repeat until the goal node is reached.

2-A* Search

Avoid expanding paths that are already expensive.
A∗ search uses an admissible heuristic, one that never overestimates the cost to reach goal
Theorem: A∗ search is optimal



If node B is the goal node from node A.
Expand the available nodes to choose from ;

From A, can move to
S 140 (cost from A to S) + 253 (cost to [the goal] from S) = 393
T 118 (cost from A to T) + 329 (est cost to B [the goal] from T) = 447
Z 75 + 374 = 449

choose the lowest total cost. which is S from A

From S, can move to
A - 646=280 (accumulated cost from A -> S -> A) +366 (est cost to B [the goal] from S)
F - 417=239+178
O - 671=291+380
R - 413=220+193

choose the lowest total cost, which is R from S

like so continue till the goal node is reached.

7 - Machine Learning

Saturday, November 1, 2025 12:21 PM

Machine Learning

- Supervised
- Unsupervised
- Reinforcement Learning

Supervised

- Regression vs. Classification
- Linear regression
- Decision trees
- Applications
- Research

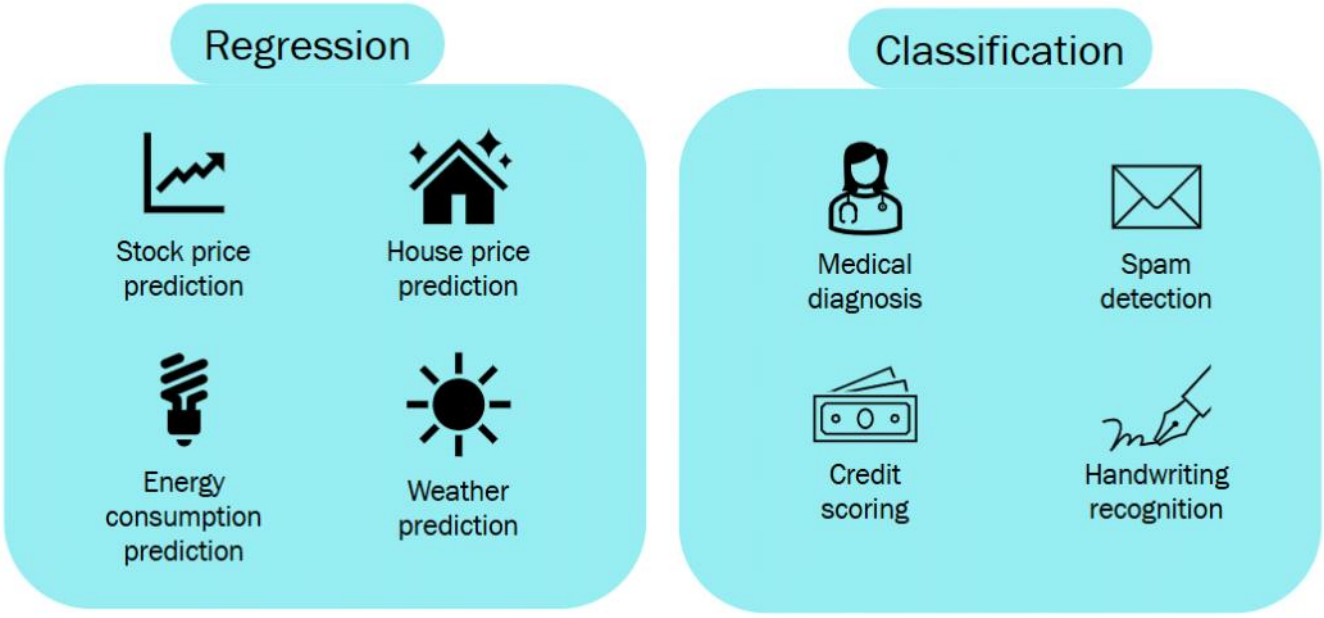
Unsupervised

- K-means
- Applications
- Research

Data

- Numerical vs. categorical
- Data split
- Accuracy metrics
- Confusion matrix

SUPERVISED LEARNING: APPLICATIONS



- Use **classification** when your output is a **label**.
- Use **regression** when your output is a **number**.

LINEAR REGRESSION

REGRESSION: LINEAR REGRESSION

- Finds the correlation between variables and enables the prediction of the continuous output variable based on one or more predictor variables.
- **Simple linear regression** (one independent variable): The model finds the linear relationship between the independent variable (x) and the dependent variable (y):

$$y = \underbrace{mx + b}_{\text{Equation of the Line}} + \varepsilon$$

Where:

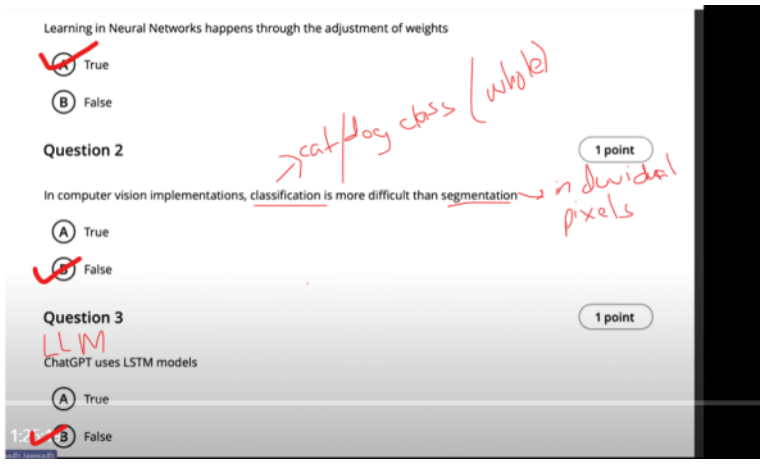
- y is the dependent variable (**output** - price)
- x is the independent variable (**input** – square feet)
- m is the slope (how the dependent variable changes with a unit change in the independent variable)
- b is the intercept (the point where the regression line intersects the y-axis, value of y when x is 0)
- ε is the error term (residual) between the predicted and actual values

- **Linear Regression** fits a line to the data.

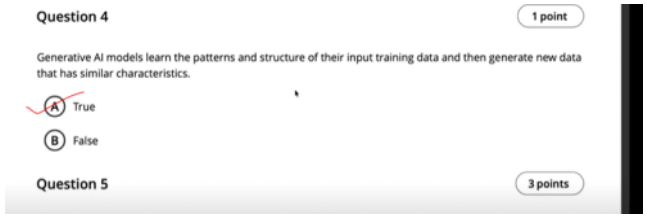
- It’s used for **predicting continuous values**.
- Works best when data shows a **linear trend**.
- Evaluation metrics include **MSE**, **RMSE**, and **R^2**.

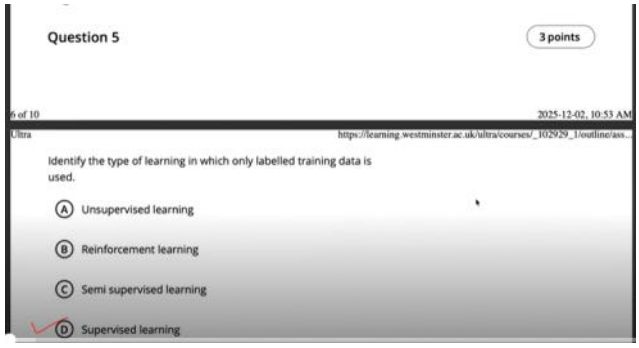
mock2

Saturday, December 6, 2025 11:51 AM

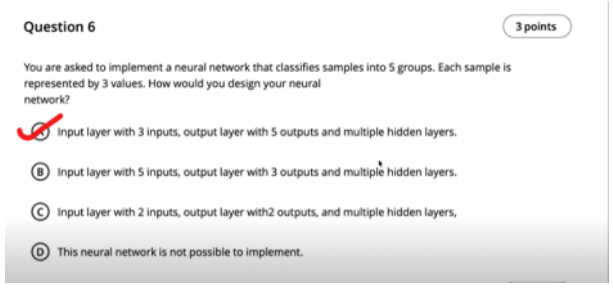


3-
False.
ChatGPT does **not** use an LSTM model.
It is built on **Transformer architecture**, which uses **self-attention**, not LSTM recurrent networks.

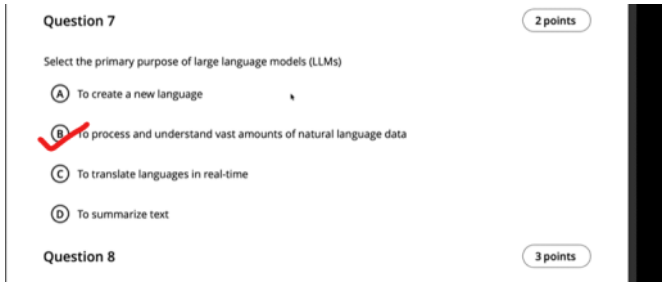




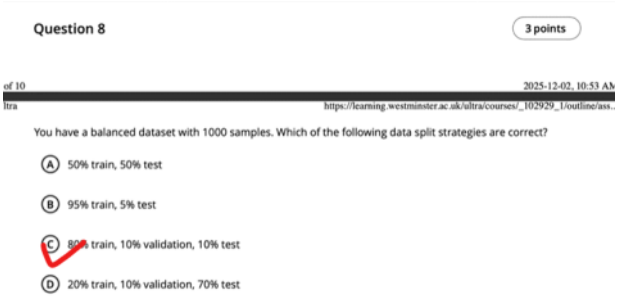
auto encoders, tranformers , generative adversarial networks



classifying into 5 groups, hence output is 5



LLM - vas amount of data



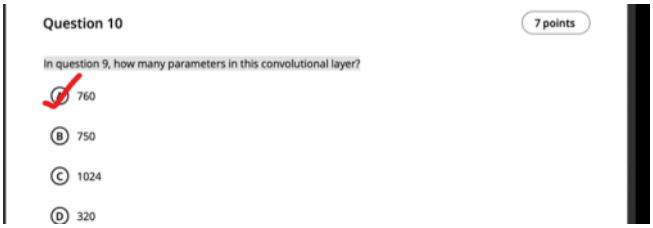
The more data u train, the higher the accuracy



32 -, 5 is the size of the filters, right? Then we are adding two multiplied by the padding size, right? Dividing it by the style value. Style value here is 1.

Right. So that is what happening here and the value we are getting is 32. So 32 by 32 and 10, right? So that is the value. Hope it is understandable. The depth is equal to number of filters, right? So the number of filters is 10.

In this case, so we have to multiply that by, uh, the number of filters.
>



weights per filters and the biases per filters and we can end that with the what we call this we can multiply it by.

The depth or the number of filters? No. So the number of filters here they have shared is 10. Yes 10 we have to multiple that. So that is the ideal value.

>